Xcelium 24.09 + VManager

Installation & Coverage-Engine Setup Guide

CentOS 7 / RHEL 7 | Cadence EDA Toolchain

Manual package extraction · IMC coverage-engine resolution

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Contents

1. **Phase 0 — Pre-flight Check**
   Confirm what is already installed and whether you need to install anything at all.
2. **Phase 1 — Transfer Packages**
   Move kits from Windows without destroying Unix permissions.
3. **Phase 2 — Install Xcelium 24.09**
   Manual extraction (faster and more reliable than IScape).
4. **Phase 3 — Install VManager**
   Provides `imc`, which Xcelium does not ship.
5. **Phase 4 — Environment Setup**
   The critical phase. Where the coverage-engine mismatch is fixed.
6. **Phase 5 — Verify End to End**
   Prove the writer and reader versions match.
7. **Troubleshooting**
   Every failure encountered during the real installation, with fixes.
8. **Key Takeaways**

> **Why this guide exists**
>
> Coverage databases (`.ucd` / `.ucm`) are versioned binary files. The viewer must belong to the same tool generation as the simulator that wrote them, and there is no upgrade path. On a machine with several Cadence installs, **IMC resolves its coverage engine from whichever simulator wins the `PATH` race** — not from the IMC binary you invoke. That single fact is the root of nearly every "serialization exception" reported here.

Phase 0 — Pre-flight Check

Run this **before** anything else. It takes two minutes and may tell you no installation is needed.

```
which -a xrun irun imc vmanager
ls /home/install/ 2>/dev/null || ls /opt/ /tools/ 2>/dev/null

xrun -version
irun -version
imc  -version

df -h /home/install

which uncompress tar zcat java
java -version
```

Install prerequisites

```
yum install -y ncompress java-1.8.0-openjdk fontconfig \
               dejavu-sans-fonts liberation-fonts
```

Interpreting the result

What you see	What it means
Simulator and IMC report the *same* major version	Matched pair already present. Stop here — nothing to install.
Simulator and IMC report *different* major versions	Continue with this guide.
`imc` not found at all	Xcelium does not ship IMC. You need VManager (Phase 3).
Multiple installs under `/home/install/`	Expect PATH-resolution conflicts. Phase 4 is mandatory.

> **Disk space**
>
> A full Xcelium install needs roughly **30 GB free** — about 12 GB of compressed packages expanding to 25 GB. Verify with `df -h` before starting.

License server sanity check

`lmstat` frequently reports a false "server down" when its own version does not match the server's `lmgrd`. Test the TCP ports directly instead — ICMP is usually firewalled, so a failing `ping` proves nothing.

```
echo $LM_LICENSE_FILE
nc -zv <license_server_ip> 5280
nc -zv <license_server_ip> 5281
```

If both ports connect, licensing is fine regardless of what `lmstat` claims.

Phase 1 — Transfer Packages from Windows

> **Do not drag and drop**
> NTFS carries no Unix permission bits. A plain copy or drag-and-drop delivers every file as `644` — nothing executable. The Cadence installer then fails on *every* component, because it shells out to helper binaries it can no longer run. Always archive first.

On Windows (PowerShell)

```
cd "F:\Cadence_Tools\downloads"
tar -czf Xcelium_2409.tar.gz Xcelium_2409
tar -czf Vmanager.tar.gz Vmanager
```

Windows 10 and later ship `tar` natively. Archiving 12 GB takes a while.

Transfer

```
scp "F:\Cadence_Tools\downloads\Xcelium_2409.tar.gz" root@<TARGET_IP>:/home/install/
scp "F:\Cadence_Tools\downloads\Vmanager.tar.gz"     root@<TARGET_IP>:/home/install/
```

WinSCP or MobaXterm work equally well provided you transfer the *archives*, not the folders.

On Linux — unpack and normalise permissions

```
cd /home/install
tar -xzf Xcelium_2409.tar.gz
tar -xzf Vmanager.tar.gz
chmod -R u+rwX,go+rX /home/install/Xcelium_2409 /home/install/Vmanager
ls /home/install/Xcelium_2409/kits/ | wc -l
```

Expect roughly **321 files** in `kits/`. Verify against the source before continuing:

```
du -sb /home/install/Xcelium_2409/kits/
```

Phase 2 — Install Xcelium 24.09

The IScape GUI installer is unreliable on older Linux distributions — broken font rendering, missing native libraries, and permission-sensitive helper binaries. The `.t.Z` packages are ordinary `tar` + `compress` archives, so **manual extraction is faster and has far fewer failure modes**.

Step 1 — Verify one archive first

```
zcat /home/install/Xcelium_2409/kits/xmcoviccDoc24.09-s001.t.Z | tar tf - | head
```

A clean file listing means the archives are intact and extraction will work.

Step 2 — Extract everything

```
mkdir -p /home/install/XCELIUM2409
cd /home/install/XCELIUM2409

nohup bash -c 'for f in /home/install/Xcelium_2409/kits/*.t.Z; do
  echo "=> $(basename $f)"
  zcat "$f" | tar xf - || echo "FAILED: $f" >> /tmp/extract_failed.log
done' > /tmp/extract.log 2>&1 &
```

Runs in the background and survives a dropped session. Monitor progress:

```
grep -c '^=>' /tmp/extract.log
du -sh /home/install/XCELIUM2409
```

Roughly 320 packages, 20–30 minutes.

Step 3 — Verify and create the `tools` symlink

IScape's configure step normally creates `tools → tools.lnx86`. Manual extraction does not, so create it yourself:

```
cat /tmp/extract_failed.log 2>/dev/null || echo "no failures"
chmod -R u+rwX,go+rX /home/install/XCELIUM2409
cd /home/install/XCELIUM2409
ln -s tools.lnx86 tools
/home/install/XCELIUM2409/tools/bin/xrun -version
```

Must print `TOOL: xrun 24.09-s001`.

> **Why the symlink matters**
>
> Every Cadence install exposes binaries at `<install>/tools/bin/`, but the real files live in `tools.lnx86/`. The binaries themselves are further symlinked (`xrun → ../inca/bin/xrun`), which `tar` preserves. Only the top-level `tools` link is missing after manual extraction.

Phase 3 — Install VManager (provides `imc`)

> **Xcelium does not ship IMC**
> Searching an Xcelium install for `imc` returns only `launch_rmi_imc.sh` — a wrapper that calls whatever `imc` it finds on `PATH`. The actual binary comes from VManager / Verisium Manager, and is itself a shell script that invokes `vmanager -metrics_only`.

Step 1 — Check whether VManager is already extracted

```
ls /home/install/Vmanager/
ls /home/install/Vmanager/kits/bin/imc 2>/dev/null
ls /home/install/Vmanager/kits/tools/bin/imc 2>/dev/null
```

If an `imc` binary already exists, skip to Phase 4 and note its exact path. If `kits/` contains only `.t.Z` files, extract as below.

Step 2 — Extract

```
mkdir -p /home/install/VMANAGER2409
cd /home/install/VMANAGER2409

for f in /home/install/Vmanager/kits/*.t.Z; do
  echo "=> $(basename $f)"
  zcat "$f" | tar xf -
done

chmod -R u+rwX,go+rX /home/install/VMANAGER2409
ln -s tools.lnx86 tools 2>/dev/null
ls tools/bin/imc
```

VManager is a single package, so this finishes in a minute or two.

Step 3 — Record the exact `imc` path

This step is essential and easy to skip. The directory layout differs depending on how VManager was installed, and Phase 4 depends on getting it right:

```
find /home/install -name imc -type f 2>/dev/null
find /home/install -name imc 2>/dev/null
```

Install method	Path to `imc`	Use in `$path`
IScape / vendor installer	`<VM>/kits/bin/imc`	`$VMANAGERHOME/bin`
Manual extraction (this guide)	`<VM>/tools/bin/imc`	`$VMANAGERHOME/tools/bin`

> **Observed in practice**
> On one machine the working path was `VMANAGERHOME=/home/install/Vmanager/kits` with `$VMANAGERHOME/bin`. On a second machine it was `VMANAGERHOME=/home/install/VMANAGER2409` with `$VMANAGERHOME/tools/bin`. Setting the wrong one produces no error — the shell simply skips the non-existent directory and silently falls through to an *older* `imc` elsewhere on `PATH`. Always `ls` for the binary before writing the config.

Step 4 — Confirm the version

```
/home/install/VMANAGER2409/tools/bin/imc      -version
/home/install/VMANAGER2409/tools/bin/vmanager -version
```

Both should report `24.09-s001`. Note that `-version` reports only the front-end; the coverage engine version appears solely when loading a database, and is verified in Phase 5.

Phase 4 — Environment Setup

This is the phase that actually fixes the coverage problem. Everything before it merely put files on disk.

> **The mechanism**
>
> `imc` is a shell script calling `vmanager -metrics_only`. VManager **detects the simulator from the environment at runtime** and loads that simulator's coverage engine. If an older Incisive or Xcelium appears earlier in `$path` — or if `LD_LIBRARY_PATH` points at its libraries — IMC loads the *old* engine and cannot read databases the new simulator wrote. The IMC binary's own version is irrelevant.

Edit the shared `cshrc` (typically `/home/install/cshrc`). If it is read-only, copy it into your project directory and source that instead — it behaves identically.

4a — Home variables

```
setenv XCELIUMHOME   /home/install/XCELIUM2409
setenv VMANAGERHOME  /home/install/VMANAGER2409
```

Use whichever VManager path you confirmed in Phase 3, Step 3.

4b — Fix `LD_LIBRARY_PATH`

Existing configurations commonly contain two consecutive assignments where the second *overwrites* the first, leaving only the old Incisive libraries:

```
setenv LD_LIBRARY_PATH  $CDS_INST_DIR/tools/lib:$CDS_INST_DIR/tools/dfII/lib
setenv LD_LIBRARY_PATH  $IUSHOME/tools/lib          <-- overwrites, points at 15.20
```

Replace with:

```
setenv LD_LIBRARY_PATH  $CDS_INST_DIR/tools/lib:$CDS_INST_DIR/tools/dfII/lib
setenv LD_LIBRARY_PATH  ${XCELIUMHOME}/tools.lnx86/lib:${LD_LIBRARY_PATH}:${IUSHOME}/tools/lib
```

Xcelium 24.09's libraries now take precedence; Incisive remains available so `irun` and AMS flows keep working.

4c — Force Xcelium and VManager to the front of `$path`

Locate the long `set path = (...)` line. **Immediately after it**, add:

```
############################################################
#####  Xcelium + VManager must precede older installs  ####
#####  so IMC resolves the matching coverage engine    ####
set path = ($XCELIUMHOME/tools/bin $XCELIUMHOME/bin \
            $XCELIUMHOME/tools.lnx86/bin \
            $VMANAGERHOME/tools/bin $VMANAGERHOME/tools.lnx86/bin \
            $VMANAGERHOME/bin $path)

#####  Fallback: uncomment ONLY if IMC still reports   ####
#####  an old engine. Warning: breaks irun / AMS.      ####
#unsetenv IUSHOME
#unsetenv AMSHOME
#setenv LD_LIBRARY_PATH ${XCELIUMHOME}/tools.lnx86/lib
```

> **Ordering rule — the most common mistake**
>
> `set path = (... $path)` *prepends*. Whichever assignment runs **last** ends up at the front of `PATH`. Placing the Xcelium line *before* the long original line silently defeats the entire fix, because the long line then pushes everything else back in front of it. **The Xcelium line must come last.**

Including all three candidate VManager subdirectories (`tools/bin` , `tools.lnx86/bin` , `bin`) is deliberate. Non-existent directories are harmlessly skipped, so the same config works across both install layouts.

4d — Reference: correct ordering

```
setenv XCELIUMHOME      /home/install/XCELIUM2409
setenv VMANAGERHOME     /home/install/VMANAGER2409
...
setenv LD_LIBRARY_PATH  $CDS_INST_DIR/tools/lib:$CDS_INST_DIR/tools/dfII/lib
setenv LD_LIBRARY_PATH  ${XCELIUMHOME}/tools.lnx86/lib:${LD_LIBRARY_PATH}:${IUSHOME}/tools/lib
...
set path = ($VMANAGERHOME/... $IUSHOME/... ... $path)     <-- original, unchanged
set path = ($XCELIUMHOME/tools/bin ... $VMANAGERHOME/tools/bin ... $path)   <-- NEW, last
#unsetenv IUSHOME                                          <-- commented fallback
...
clear
echo "Welcome to Cadence Tools Suite"
```

Phase 5 — Verify End to End

Step 1 — Environment resolution

```
tcsh
source /home/install/cshrc

which xrun
xrun -version
which imc
imc -version
echo $path | tr ' ' '\n' | head -6
echo $LD_LIBRARY_PATH
```

Check	Expected
`which xrun`	`/home/install/XCELIUM2409/tools/bin/xrun`
`xrun -version`	`24.09-s001`
`which imc`	Under your `VMANAGERHOME`, *not* under an Incisive install
`imc -version`	`24.09-s001`
First entries of `$path`	XCELIUM2409, then VMANAGER
`LD_LIBRARY_PATH`	Begins with `/home/install/XCELIUM2409/tools.lnx86/lib`

> **Do not paste inline comments into tcsh**
> tcsh does not strip `# comments` from interactive command lines. Pasting `xrun -version  # must be 24.09` feeds the comment words to the command as arguments. Paste the command only.

Step 2 — Round-trip a real database

```
cd <project>/sim/cadence_sim
make clean
rm -rf cov_work
make compile
make simulate test=<your_test>

ls -la cov_work/scope/
imc -execcmd "load cov_work/scope/<your_test>; exit"
```

Success looks like this — both versions identical:

```
IMC(64): 24.09-s001: (c) Copyright ... Cadence Design Systems Inc
         - Simulator version: 24.09-s001 - Coverage engine version: 24.09-s001
Starting batch mode
*I,RUNLD: Successfully loaded run /.../cov_work/scope/<your_test>.
```

Failure looks like this — versions differ, so return to Phase 4:

```
*I,imc.tool.cov.version.diff: "There is a difference between the IMC tool
    version 24.09 and the coverage version 15.20-s086"
IMC(64): 24.09-s001: ... - Coverage engine version: 15.20-s086
*E,DBERR: Error loading database file .../icc_xxx.ucm: serialization exception.
*E,INVDT: Valid data file not found for run directory ...
```

Step 3 — Generate the report

```
imc -execcmd "load cov_work/scope/<your_test>; report_metrics -out cov_report; exit"
firefox cov_report/index.html &
```

Or browse interactively:

```
imc -gui -execcmd "load cov_work/scope/<your_test>" &
```

> **Correct load syntax**
> Use `-execcmd "load <path>"`. The `-load` flag cannot handle the split layout where the `.ucm` model sits at `cov_work/scope/` while the `.ucd` data sits at `cov_work/scope/<test>/`. The command `load_test` does not exist in 24.09 — it returns `invalid command name`. Prefer `report_metrics` over the deprecated `report -html`.

Makefile integration

```
merge_cov_report:
        @test -d cov_work || { echo "ERROR: cov_work does not exist"; exit 2; }
        @rm -rf merged_cov_html_report merged_coverage
        $(IMC) -execcmd "merge cov_work/scope/*/ -out merged_coverage; \
                         load merged_coverage; \
                         report_metrics -out merged_cov_html_report; exit"
        @echo "Coverage report: merged_cov_html_report/index.html"
```

Troubleshooting

`serialization exception` when loading a database

```
*E,DBERR: Error loading database file .../icc_xxx.ucm: serialization exception.
```

Cause. IMC's coverage engine version differs from the simulator that wrote the database.

Diagnose. Read the banner — the engine version is printed on load:

```
imc -execcmd "load cov_work/scope/<test>; exit" |& head -3
```

If `Coverage engine version` differs from `xrun -version`, that is the fault.

Fix. Phase 4. If reordering is insufficient, uncomment the fallback block and re-source.

`which imc` finds the wrong install

The `$VMANAGERHOME` subdirectory you added to `$path` does not exist, so the shell skipped it and fell through to an older `imc`. Non-existent path entries produce no warning.

```
which -a imc
ls $VMANAGERHOME/bin/imc
ls $VMANAGERHOME/tools/bin/imc
```

Correct `VMANAGERHOME` or the subdirectory, then re-source.

IMC's engine stays on the old version after Phase 4

Test with a minimal environment to confirm the diagnosis:

```
setenv PATH /home/install/XCELIUM2409/tools/bin:/home/install/XCELIUM2409/bin:\
/home/install/VMANAGER2409/tools/bin:/usr/bin:/bin
unsetenv IUSHOME
unsetenv AMSHOME
unsetenv LD_LIBRARY_PATH
setenv XCELIUMHOME  /home/install/XCELIUM2409
setenv CDS_INST_DIR /home/install/XCELIUM2409
rehash
imc -execcmd "load cov_work/scope/<test>; exit"
```

If this works, the ordering in your `cshrc` is still wrong. If it does not, IMC is reading a variable not yet accounted for — inspect with `env | grep -iE "cds|xcel|inci|vmgr"`.

IScape: "Install from CDROM(s) Errors" — every component fails

Cause. Execute bits stripped by an NTFS copy. IScape shells out to helper binaries (`vtar`, `vload`, `cdsLoad`) that are no longer executable.

```
chmod -R u+rwX,go+rX /home/install/Xcelium_2409/
find /home/install/Xcelium_2409/install/bin.lnx86 -type f -exec chmod +x {} \;
yum install -y ncompress
```

Or bypass IScape entirely with the Phase 2 manual extraction.

IScape GUI: unreadable fonts

```
yum install -y fontconfig dejavu-sans-fonts liberation-fonts
fc-cache -f
java -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true \
     -Dsun.java2d.xrender=false -Dsun.java2d.uiScale=1.5 \
     -jar InstallScape.jar &
```

Paste as a single line. Backslash continuations break when a terminal adds trailing whitespace.

IScape: `libnativemethods_lnx_64.so` warning

Harmless. That library only opens a web browser for README links. Dismiss and continue.

IScape: bogus "insufficient disk space"

IScape's own space check misreports on some filesystems. Confirm with `df -h`; if space is genuinely available, click OK and proceed.

Shell syntax quick reference

Most "command not found" and "ambiguous output redirect" errors are bash syntax typed into tcsh or the reverse.

Task	bash	tcsh
Set environment variable	`export VAR=val`	`setenv VAR val`
Discard stderr	`2>/dev/null`	`>& /dev/null`
Capture both streams	`&> file`	`>& file`
Pipe both streams	`2>&1 |`	`|&`
Rebuild command cache	`hash -r`	`rehash`
Command substitution	`$(cmd)`	`` `cmd` ``

Verifying an archive before blaming the transfer

```
ls /home/install/Xcelium_2409/kits/ | wc -l
du -sb /home/install/Xcelium_2409/kits/
zcat /home/install/Xcelium_2409/kits/<any>.t.Z | tar tf - | head
```

Directory entries such as `data/` and `doc/` inside `kits/` are leftovers from a previous extraction run inside that folder. They are harmless, and they explain small file-count and size discrepancies against the Windows source.

Key Takeaways

Principle	Detail
Check before installing	Phase 0 takes two minutes. A machine with a matched simulator/viewer pair needs no work at all.
Archive before transferring	NTFS has no execute bit. Copying a Cadence tree directly from Windows renders every binary non-executable and produces installer failures that look like corruption.
Manual extraction beats IScape	`.t.Z` files are `tar + compress`. Extracting them directly avoids the GUI's font, native-library, and permission failure modes. Only the `tools` symlink must be recreated by hand.
Coverage databases are version-locked	The viewer must belong to the same generation as the writer. No upgrader exists, and `serialization exception` is the signature of a mismatch — not of a corrupt file.
IMC follows the PATH	The single most important fact in this document. `which imc` reporting 24.09 means nothing if an older simulator wins the `PATH` race. Trust only the `Coverage engine version` line printed when a database is loaded.
Prepend order is reversed	`set path = (... $path)` prepends, so the *last* assignment wins the front position. The new-tool line must come after the long original one.
Verify the VManager layout	`bin/imc` and `tools/bin/imc` both occur in the wild. The wrong choice fails silently by falling through to an older install.

Minimum viable verification

Whatever else changes, this sequence proves the toolchain is coherent:

```
xrun -version
imc  -version
imc -execcmd "load cov_work/scope/<test>; exit"
```

If all three agree on the version and the load succeeds, the installation is correct.

> **Applies to other vendors**
> The same writer/reader generation rule governs Synopsys (`vcs` writes `.vdb` ; `urg` and Verdi read it) and Siemens/Mentor (`qrun` and `vsim` write `.ucdb` ; `vcover` and Questa read it). When no matched pair is available, the Accellera UCIS interchange format is the fallback bridge — lossy, but sometimes the only route.